

XYZ Bank Spring Boot Assignment - Completion Report

This PDF contains the completed functional requirements, validations, security features, APIs, and technical implementation details of the XYZ Bank Spring Boot Assignment project.

Functional Requirements Completed

- Customer registration implemented
- Login using username and auto-generated password implemented
- JWT token authentication implemented
- Async ID document upload implemented
- File size validation implemented
- File format validation implemented
- Unique 16-character ABNA account number generation implemented
- Country restriction validation implemented
- 18+ age validation implemented
- Account overview API implemented
- Document download API implemented

API Endpoints

- POST /register
- POST /login
- GET /overview
- POST /documents/upload/{customerId}
- GET /documents/download/{id}

Technical Requirements Completed

- Spring Boot REST architecture
- Java 21 compatibility
- MySQL integration
- Lombok integration
- Swagger/OpenAPI integration
- JWT security
- Spring Security
- Generic exception handling
- Async processing

Validation Features

- Email validation
- Required field validation
- Age validation
- Country validation
- Duplicate email validation
- Duplicate username validation
- File empty validation
- File size validation
- File type validation

Security Features

- JWT token generation
- JWT token validation
- JWT authentication filter
- BCrypt password encryption
- Protected endpoints
- AuthenticationProvider
- UserDetailsService

Database Features

- Customer entity
- Account entity
- Document entity
- JPA relationships
- MySQL schema generation

JUnit Testing

- Service layer tests
- Controller layer tests
- Mockito tests
- MockMvc tests

Final Status: The project implementation satisfies the assignment requirements, including REST APIs, JWT authentication, MySQL integration, validation handling, Swagger documentation, async file upload, and JUnit testing.